

# American and Bermudan Option Pricing in Multifactor Stochastic Volatility Models

Half-PhD Seminar

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# One Numerical Idea Across Three Papers

One common numerical idea connects the three papers: use the structure of the stochastic-volatility model to improve either path simulation or continuation-value approximation.

## Paper A

### AES

Heston and double Heston. CIR variance sampling is used to reduce Euler discretization bias in early-exercise pricing.

## Paper B

### AEMS / AEMS-SOR

Gatheral DMR. AES-type sampling is combined with a Milstein correction for a model with stochastic long-run variance.

## Paper C

### Hybrid LSMC-PDE

GDMR Bermudan pricing. The stock-price direction is handled by a conditional FFT/PDE step and regression is performed over variance variables.

# Bermudan and American Pricing under Stochastic Volatility

## Pricing problem

For a put payoff  $h(S_t) = (K - S_t)^+$ ,  
Bermudan/American pricing requires

$$V_{t_k} = \max\{h(S_{t_k}), C(t_k, Y_{t_k})\},$$

where

$$C(t_k, Y_{t_k}) = e^{-r\Delta t} \mathbb{E}^{\mathbb{Q}}[V_{t_{k+1}} \mid Y_{t_k}].$$

## Computational challenge

- State variables include asset price and one or more variance factors.
- Simulation bias affects exercise decisions.
- Continuation values are conditional expectations, not explicit formulas.

The numerical price is driven by two decisions: how accurately paths are simulated, and how accurately continuation values are approximated.

# Two Numerical Difficulties

## 1. Variance simulation

Euler-type schemes can create negative variance approximations and introduce discretization bias.

$$dv_t = \kappa(\theta - v_t) dt + \xi\sqrt{v_t} dW_t.$$

For CIR-type variance factors, the next variance value can be sampled from its known noncentral chi-square distribution.

## 2. Continuation regression

In multifactor stochastic volatility,

$$C(t, Y_t) = \mathbb{E}^{\mathbb{Q}}[\text{discounted future value} \mid Y_t].$$

Plain LSMC must approximate this high-dimensional object from simulated paths.

Papers A and B focus on the simulation of variance paths. Paper C focuses on the approximation of continuation values.

# Model Hierarchy

## Heston

$$dS_t = rS_t dt + \sqrt{v_t} S_t dW_t^{(1)}$$

$$dv_t = \kappa(\theta - v_t) dt + \xi \sqrt{v_t} dW_t^{(2)}$$

## Double Heston

$$dS_t = rS_t dt + S_t \sum_{i=1}^2 \sqrt{v_t^{(i)}} dW_t^{(i)}$$

$$dv_t^{(i)} = \kappa_i(\bar{v}_i - v_t^{(i)}) dt + \gamma_i \sqrt{v_t^{(i)}} dW_t^{(i+2)}, \quad i = 1, 2.$$

## Gatheral DMR/GDMR

$$dS_t = rS_t dt + S_t \sqrt{v_t} dW_t^{(1)}$$

$$dv_t = \kappa_1(v_t' - v_t) dt + \xi_1 v_t^{\delta_1} dW_t^{(2)}$$

$$dv_t' = \kappa_2(\theta - v_t') dt + \xi_2 (v_t')^{\delta_2} dW_t^{(3)}$$

Paper B uses the square-root DMR case,  $\delta_1 = \delta_2 = \frac{1}{2}$ . Paper C treats the CEV-type GDMR extension.

# Paper A: AES for Heston-Type Models

Paper A extends the Almost-Exact Simulation scheme to Bermudan and American put options under Heston and double Heston models.

## The idea

- Sample each CIR variance factor from its noncentral chi-square distribution.
- Use the sampled variance values in the log-price update.
- Avoid the negative variance values that may appear in standard Euler discretization.

## Terminology

AES is not fully exact. It is almost exact because remaining stochastic integrals in the log-price update are still approximated.

## Role in early-exercise pricing

Paper A reports higher accuracy and computational efficiency when the number of simulation steps matches the exercise dates for Bermudan options.

# Paper A Derivation: Double Heston AES One-Step Update

## 1. Double Heston log-price form

Applying Itô's lemma to  $X_t := \ln S_t$  and a Cholesky decomposition gives

$$\begin{aligned} dX_t &= \left( r - \frac{1}{2}(\nu_t^1 + \nu_t^2) \right) dt + \sqrt{\nu_t^1} \left( \rho_{1,3} \widetilde{dW}_t^3 + \sqrt{1 - \rho_{1,3}^2} \widetilde{dW}_t^1 \right) \\ &\quad + \sqrt{\nu_t^2} \left( \rho_{2,4} \widetilde{dW}_t^4 + \sqrt{1 - \rho_{2,4}^2} \widetilde{dW}_t^2 \right), \\ d\nu_t^1 &= \kappa_1(\bar{\nu}_1 - \nu_t^1) dt + \gamma_{\nu_1} \sqrt{\nu_t^1} \widetilde{dW}_t^3, \\ d\nu_t^2 &= \kappa_2(\bar{\nu}_2 - \nu_t^2) dt + \gamma_{\nu_2} \sqrt{\nu_t^2} \widetilde{dW}_t^4. \end{aligned}$$

The four  $\widetilde{W}$ -increments are independent.

## 2. Sample the two variance factors

$$\begin{aligned} \nu_{j+1}^{(i)} &= \bar{c}^{(i)} \chi_{\delta^{(i)}}^2 \left( \bar{\kappa}^{(i)} \right), \quad i = 1, 2, \\ \bar{c}^{(i)} &= \frac{\gamma_{\nu_i}^2}{4\kappa_i} (1 - e^{-\kappa_i \Delta t}), \quad \bar{\kappa}^{(i)} = \frac{4\kappa_i \nu_j^{(i)} e^{-\kappa_i \Delta t}}{\gamma_{\nu_i}^2 (1 - e^{-\kappa_i \Delta t})}, \\ \delta^{(i)} &= \frac{4\kappa_i \bar{\nu}_i}{\gamma_{\nu_i}^2}. \end{aligned}$$

## 3. Constants used in the log-price update

$$\begin{aligned} c_0 &= \left( r - \frac{\rho_{1,3}\kappa_1\bar{\nu}_1}{\gamma_{\nu_1}} - \frac{\rho_{2,4}\kappa_2\bar{\nu}_2}{\gamma_{\nu_2}} \right) \Delta t, \\ c_1 &= \left( \frac{\rho_{1,3}\kappa_1}{\gamma_{\nu_1}} - \frac{1}{2} \right) \Delta t - \frac{\rho_{1,3}}{\gamma_{\nu_1}}, \quad c_2 = \left( \frac{\rho_{2,4}\kappa_2}{\gamma_{\nu_2}} - \frac{1}{2} \right) \Delta t - \frac{\rho_{2,4}}{\gamma_{\nu_2}}, \\ c_3 &= \frac{\rho_{1,3}}{\gamma_{\nu_1}}, \quad c_4 = \frac{\rho_{2,4}}{\gamma_{\nu_2}}, \quad c_5 = (1 - \rho_{1,3}^2) \Delta t, \quad c_6 = (1 - \rho_{2,4}^2) \Delta t. \end{aligned}$$

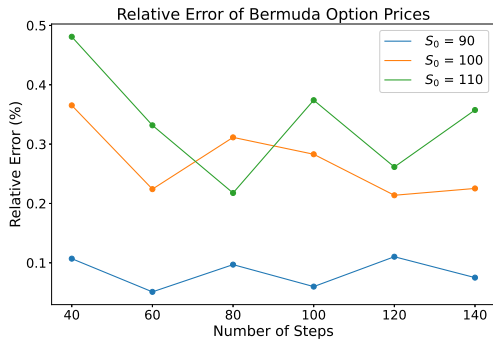
## 4. Almost-exact asset update

$$\begin{aligned} X_{j+1} &= X_j + c_0 + c_1 \nu_j^{(1)} + c_2 \nu_j^{(2)} + c_3 \nu_{j+1}^{(1)} + c_4 \nu_{j+1}^{(2)} \\ &\quad + \sqrt{c_5 \nu_j^{(1)}} (W_{j+1}^{(1)} - W_j^{(1)}) + \sqrt{c_6 \nu_j^{(2)}} (W_{j+1}^{(2)} - W_j^{(2)}). \end{aligned}$$

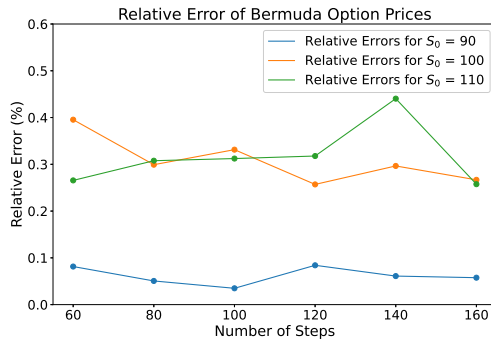
Finally set  $S_j = \exp(X_j)$ .

# Paper A: Bermudan Heston Numerical Results

## 40 exercise dates



## 60 exercise dates



With simulation steps matched to exercise dates, AES yields low relative errors, especially for in-the-money and at-the-money puts.



# Paper A: American Put Experiment

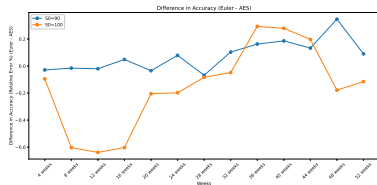
## Experiment

AES for Double Heston American put with  $M = 12$  simulation steps and three test cases. The comparison is between AES and Euler with the same time-steps, and reference values.

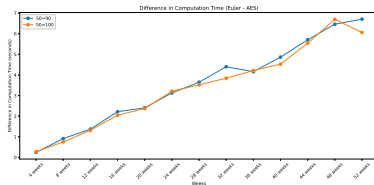
| Method          | Case 1 | Case 2 | Case 3 |
|-----------------|--------|--------|--------|
| AES, $M = 12$   | 6.992  | 9.635  | 12.676 |
| Euler, $M = 12$ | 5.116  | 7.611  | 10.608 |
| Reference       | 6.887  | 9.504  | 12.520 |

At the same small number of simulation steps, AES is close to the reference values, while Euler is materially below the reference values in all three cases.

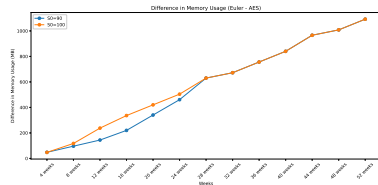
# Paper A: Efficiency Comparison



Accuracy difference.



Computation time.



Memory usage.

AES is useful when Euler requires additional time steps to reach comparable accuracy; in these cases AES can reduce computation time and memory use.

## Paper A: Double Heston American Put Results

**Double Heston American put experiment with  $M = 12$  simulation steps**

| Method          | Case 1 | Case 2 | Case 3 |
|-----------------|--------|--------|--------|
| AES, $M = 12$   | 6.992  | 9.635  | 12.676 |
| Euler, $M = 12$ | 5.116  | 7.611  | 10.608 |
| Reference       | 6.887  | 9.504  | 12.520 |

In all three cases, AES is close to the reference value, while Euler is visibly below the reference value at the same small number of time steps.

# Paper A: Contribution and Limitations

## Strengths

- The AES scheme is extended to Bermudan and American put options.
- The double Heston version is derived analytically.
- The strongest numerical evidence appears for in-the-money and at-the-money options.

## Limitations

- Out-of-the-money cases with few exercise opportunities are less uniform.
- Heston and double Heston have CIR variance factors, which makes AES naturally applicable.
- Better path simulation does not remove the continuation-value approximation problem.

Paper B moves to Gatheral's double mean-reverting model, where the long-run variance level is itself stochastic.

## Paper B: Gatheral DMR as a Non-Affine Extension

Paper B moves from Heston-type affine structure to Gatheral DMR, where the stochastic long-run variance improves flexibility but makes simulation less direct.

$$\begin{aligned}dS_t &= rS_t dt + \sqrt{v_t} S_t d\widetilde{W}_t^{(1)}, \\dv_t &= \kappa_1(v'_t - v_t) dt + \xi_1 \sqrt{v_t} d\widetilde{W}_t^{(2)}, \\dv'_t &= \kappa_2(\theta - v'_t) dt + \xi_2 \sqrt{v'_t} d\widetilde{W}_t^{(3)}.\end{aligned}$$

### Numerical implication

The long-run variance  $v'$  has a CIR structure, but the current variance  $v$  mean-reverts to the stochastic level  $v'$ . Therefore the two variance equations cannot be simulated in one closed-form exact step as in the Heston case.

## Paper B Method: AEMS and AEMS-SOR

### AEMS

- Extend the Paper A AES idea by using exact CIR sampling where possible.
- Sample the long-run variance  $v'$  from its CIR distribution.
- Update  $v$  with a Milstein-style correction.
- Use the correlation structure in the log-price update.

### AEMS-SOR

- Adds a second-order refinement.
- Aims to improve weak convergence.
- Its numerical advantage depends on payoff, maturity, and moneyness.

AEMS is a mixed scheme: exact sampling is used for  $v'$ , while  $v$  and the log price are updated by approximation formulas.

# Paper B Derivation: AEMS Correlation Decomposition

The Gatheral DMR model is first rewritten with a Cholesky decomposition so that the correlated Brownian motions are expressed through independent Brownian motions.

## 1. Integral representation

$$X_{t+\Delta t} = X_t + \int_t^{t+\Delta t} \left(r - \frac{1}{2}v_u\right) du + \rho^* \int_t^{t+\Delta t} \sqrt{v_u} dW_u^{(1)} \\ + \frac{\rho_{12} - \rho_{13}\rho_{23}}{\sqrt{1 - \rho_{23}^2}} \int_t^{t+\Delta t} \sqrt{v_u} dW_u^{(2)} + \rho_{13} \int_t^{t+\Delta t} \sqrt{v_u} dW_u^{(3)},$$

$$v_{t+\Delta t} = v_t + \kappa_1 \int_t^{t+\Delta t} (v'_u - v_u) du + \xi_1 \sqrt{1 - \rho_{23}^2} \int_t^{t+\Delta t} \sqrt{v_u} dW_u^{(2)} \\ + \xi_1 \rho_{23} \int_t^{t+\Delta t} \sqrt{v_u} dW_u^{(3)},$$

$$v'_{t+\Delta t} = v'_t + \kappa_2 \int_t^{t+\Delta t} (\theta - v'_u) du + \xi_2 \int_t^{t+\Delta t} \sqrt{v'_u} dW_u^{(3)}.$$

$$\rho^* = \sqrt{1 - \rho_{13}^2 - \frac{(\rho_{12} - \rho_{13}\rho_{23})^2}{1 - \rho_{23}^2}}.$$

## 2. Isolate the shared Brownian part of $v$

$$v_{t+\Delta t} = v_t + \kappa_1 \int_t^{t+\Delta t} (v'_u - v_u) du \\ + \xi_1 \rho_{23} \left( \frac{\sqrt{1 - \rho_{23}^2}}{\rho_{23}} \int_t^{t+\Delta t} \sqrt{v_u} dW_u^{(2)} + \int_t^{t+\Delta t} \sqrt{v_u} dW_u^{(3)} \right).$$

## 3. AEMS final one-step form

$$X_{t+\Delta t} = X_t + \left(r - \frac{1}{2}v_t\right) \Delta t + \rho^* \sqrt{v_t} \Delta W^{(1)} + \rho^{**} \sqrt{v_t} \Delta W^{(2)} \\ + \rho_{13} \left( \frac{v_{t+\Delta t} - v_t - \kappa_1(v'_t - v_t)\Delta t}{\xi_1 \rho_{23}} \right),$$

$$v_{t+\Delta t} = v_t + \kappa_1(v'_t - v_t)\Delta t + \xi_1 \sqrt{v_t} \Delta t Z_v + \frac{1}{4} \xi_1^2 \Delta t (Z_v^2 - 1), \\ v'_{t+\Delta t} = \bar{c}(t + \Delta t, t) \cdot \chi^2(\delta, \bar{\kappa}(t + \Delta t, t)),$$

$$\rho^{**} = \frac{\rho_{12} - \rho_{13}\rho_{23}}{\sqrt{1 - \rho_{23}^2}} - \rho_{13} \frac{\sqrt{1 - \rho_{23}^2}}{\rho_{23}}.$$

# Paper B Derivation: AEMS-SOR Second-Order Correction

## 1. General second-order scheme

The second-order method was introduced by Milstein and later refined by Glasserman. For

$$dX_t = a(X_t) dt + b(X_t) dW_t,$$

$$\begin{aligned}\widehat{X}_{t+\Delta t} = & \widehat{X}_t + a\Delta t + b\Delta W + (ab' + \tfrac{1}{2}b^2b'')(\Delta W \Delta t - \Delta I) \\ & + a'b\Delta I + \tfrac{1}{2}bb'[(\Delta W)^2 - \Delta t] \\ & + (aa' + \tfrac{1}{2}b^2a'')\tfrac{1}{2}(\Delta t)^2.\end{aligned}$$

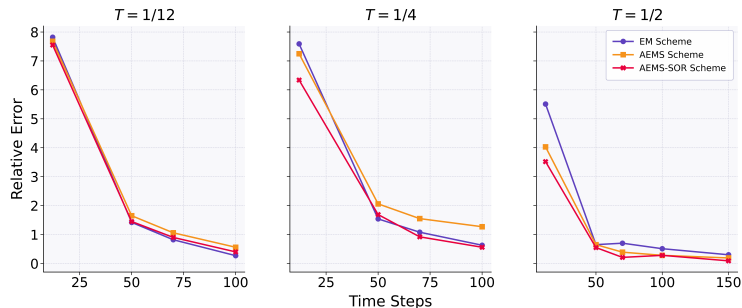
## 2. Applied to the Gatheral DMR variance equation

$$\begin{aligned}v_{t+\Delta t} = & v_t + \kappa_1(v'_t - v_t)\Delta t + \xi_1\sqrt{v_t}\Delta W^{(2)} \\ & + \left((v'_t - v_t)\frac{\xi_1}{2\sqrt{v_t}} - \frac{1}{8}\xi_1^3v_t^{-1/2}\right)\left[\Delta W^{(2)}\Delta t - \Delta I\right] \\ & + \kappa_1(v'_t - 1)\xi_1\sqrt{v_t}\Delta I + \frac{1}{4}\xi_1^2\left[(\Delta W^{(2)})^2 - \Delta t\right] \\ & + \frac{1}{2}\kappa_1^2(v'_t - v_t)(v'_t - 1)(\Delta t)^2.\end{aligned}$$

AEMS-SOR adds second-order correction terms to the AEMS variance update; this is why the scheme is more complex but can improve weak convergence in some tests.



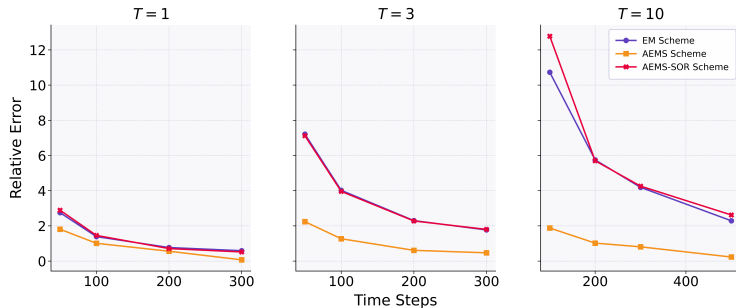
## Paper B: American Vanilla Put Results



### Interpretation

For ATM American vanilla puts, AEMS and AEMS-SOR are competitive with EM and often reduce relative error as the time grid is refined.

## Paper B: Forward Volatility Swap Result



### Benchmark setting

The forward volatility swap comparison uses an analytical benchmark, so the relative-error interpretation is stronger than in most early-exercise experiments.

### Interpretation

AEMS performs well for this variance-sensitive payoff; AEMS-SOR is not uniformly best despite its higher-order correction.

## Paper B: Basket Option Stability



### Interpretation

The basket option experiment illustrates stability across repeated runs: EM fluctuates more, while AEMS and AEMS-SOR are visibly steadier.

## Paper B: Contribution and Limitations

### Contribution

- Develops AEMS for Gatheral DMR by combining AES-type variance sampling with a Milstein correction.
- Introduces AEMS-SOR as a second-order refinement.
- Tests vanilla, barrier, forward volatility swap, and basket-style payoffs.

### Limitations

- AEMS-SOR is not the most accurate scheme in every payoff and moneyness case.
- Most option benchmarks are numerical, not analytical.
- Improved path simulation does not remove the high-dimensional LSMC regression problem.

Paper C keeps the early-exercise setting and changes the numerical focus to continuation-value approximation.

## Paper C: Hybrid LSMC-PDE under GDMR

### GDMR model

$$dS_t = rS_t dt + S_t \sqrt{v_t} dW_t^{(1)},$$

$$dv_t = \kappa_1(v'_t - v_t) dt + \xi_1 v_t^{\delta_1} dW_t^{(2)},$$

$$dv'_t = \kappa_2(\theta - v'_t) dt + \xi_2 (v'_t)^{\delta_2} dW_t^{(3)}.$$

The variance equations do not involve  $S_t$ . This one-way coupling is the main structure used by the hybrid method.

### Plain LSMC

The continuation value is regressed on the full state: asset level  $S$ , current variance  $v$ , and long-run variance  $w = v'$ .

### Hybrid LSMC-PDE

Simulate  $(v, w)$ , handle the stock-price direction on a log-price grid by conditional FFT/PDE, and regress only over  $(v, w)$ .

The goal is not to change the Bermudan recursion; it is to make the continuation-value approximation easier.

# Paper C: Hybrid Decomposition of the Continuation Value

$$U_M(s, v, w) = h_M(s), \quad U_n(s, v, w) = \max\{h_n(s), C_n(s, v, w)\}.$$

$$C_n(s, v, w) = e^{-r\Delta t_n} \mathbb{E} \left[ U_{n+1}(S_{t_{n+1}}, v_{t_{n+1}}, v'_{t_{n+1}}) \mid S_{t_n} = s, v_{t_n} = v, v'_{t_n} = w \right].$$

## 1. Inner conditional step

Simulate the variance Brownian drivers over  $[t_n, t_{n+1}]$  and condition on the enlarged information  $H_n$ . The variance segment and the correlated part of the asset Brownian motion are then fixed, leaving only the orthogonal asset Brownian component:

$$\bar{C}_n^F(s; H_n) = e^{-r\Delta t_n} \mathbb{E} \left[ F(S_{t_{n+1}}, v_{t_{n+1}}, v'_{t_{n+1}}) \mid S_{t_n} = s, H_n \right].$$

## 2. Outer regression step

The true continuation value is recovered by averaging the pathwise conditional surface over the current variance state:

$$C_n^F(s, v, w) = \mathbb{E} \left[ \bar{C}_n^F(s; H_n) \mid v_{t_n} = v, v'_{t_n} = w \right].$$

In the algorithm, this outer expectation is approximated by least-squares regression over  $(v, w)$ , while  $s$  is carried by the log-price grid.

The hybrid split changes where approximation happens: PDE/FFT handles the stock-price direction path by path, and regression handles only the variance-state dependence.

# Paper C: Hybrid LSMC-PDE Algorithm

## 1. Brownian projection

$$dW_t^{(1)} = \beta_2 dW_t^{(2)} + \beta_3 dW_t^{(3)} + \sigma_{\perp} dB_t^{\perp}.$$

Conditioning on the variance Brownian drivers leaves only the residual Brownian motion  $B^{\perp}$  in the asset direction.

## 2. Path statistics

$$Z_n^j = \sum \sqrt{\bar{v}_{\ell}^j} (\beta_2 \Delta W_{\ell}^{(2),j} + \beta_3 \Delta W_{\ell}^{(3),j}),$$
$$I_n^j = \sum (r - \frac{1}{2} \bar{v}_{\ell}^j) \Delta \tau_{\ell}, \quad B_n^j = \sigma_{\perp}^2 \sum \bar{v}_{\ell}^j \Delta \tau_{\ell}.$$

## 3. Conditional FFT step

$$u_n^j \approx \text{FFT}^{-1} \left( \text{FFT}[g_{n+1}^j] \exp\{i\omega Z_n^j + i\omega I_n^j - \frac{1}{2}\omega^2 B_n^j\} \right).$$

The terminal grid is not shifted first; the shift enters through the Fourier multiplier.

## 4. Regression over variance variables

$$\widehat{C}_n^N(s_i, v, w) = a_n^N(s_i) \cdot \phi(v, w), \quad w = v'.$$

The asset coordinate is carried by the grid  $s_i$ ; the regression learns only  $(v, w)$ .

## Paper C: Backward Hybrid Recursion

### Backward loop

- ① Initialize  $\widehat{V}_M^N(s_i, v, w) = h_M(s_i)$ .
- ② For  $n = M - 1, \dots, 1$ , compute pathwise pre-surfaces  $\widehat{C}_n^j(s_i)$  using the FFT step.
- ③ At each  $s_i$ , regress  $\widehat{C}_n^j(s_i)$  on  $\phi(v_n^j, w_n^j)$ .
- ④ Set

$$\widehat{V}_n^N(s_i, v, w) = \max\{h_n(s_i), \widehat{C}_n^N(s_i, v, w)\}.$$

### Time zero

No regression is needed at  $t_0$ . Average the first-step pre-surfaces:

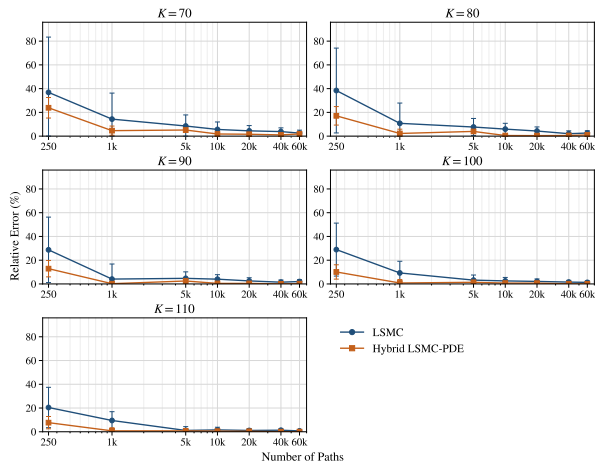
$$\widehat{C}_0^N(s_i) = \frac{1}{N} \sum_{j=1}^N \widehat{C}_0^j(s_i).$$

Then interpolate at  $S_0$  and compute

$$\widehat{V}_0^N(S_0) = \max\{h_0(S_0), \widehat{C}_0^N(S_0)\}.$$



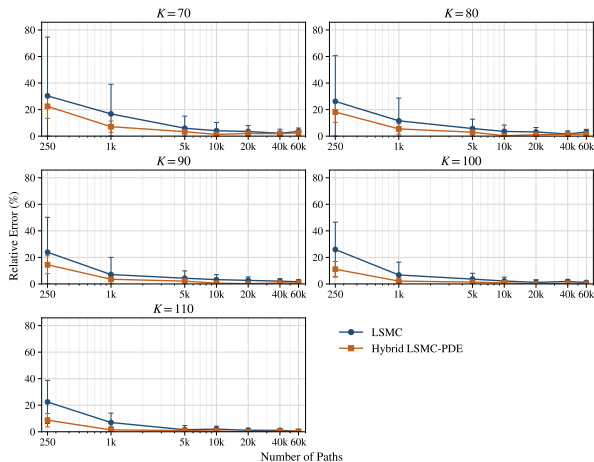
# Paper C: Path-Budget Sensitivity



## Interpretation

At fixed 60 Euler steps, the Hybrid LSMC-PDE estimator gives visibly lower relative errors in the low- and moderate-path regimes.

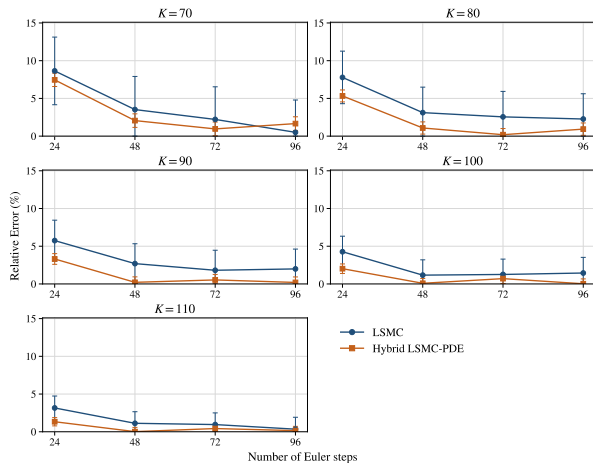
# Paper C: Path-Budget Sensitivity at 48 Euler Steps



## Interpretation

The 48-step experiment shows the same pattern: the hybrid estimator is most useful when the path budget is small or moderate.

# Paper C: Time-Step Sensitivity



## Interpretation

With 20,000 paths, the Hybrid LSMC-PDE estimator is usually centered at lower relative errors across the tested Euler step counts. The conditional PDE step reduces the dimension of the continuation regression.

## Future work

| Paper | Summary                                                                                     | Role in the thesis                                                       |
|-------|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| A     | AES is extended to Bermudan and American put pricing under Heston and double Heston models. | Improves simulation when CIR variance factors are available.             |
| B     | AEMS and AEMS-SOR are developed for the Gatheral DMR model and tested on several payoffs.   | Extends the simulation study to a stochastic long-run variance setting.  |
| C     | Hybrid LSMC-PDE is adapted to GDMR Bermudan pricing.                                        | Reduces the continuation-value regression from $(S, v, w)$ to $(v, w)$ . |

### Main conclusion

The three papers together develop simulation and continuation-value approximation tools for early-exercise pricing under multifactor stochastic volatility models.

### Future directions

Test more calibrated parameter sets; add low-biased and dual benchmarks; study more payoff types; and combine improved variance simulation with the Hybrid LSMC-PDE continuation approximation.

**Thank you for your attention**